Ilyas Mansurov

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Education

Kazakh-British Technical University Almaty, Kazakhstan
B.Sc. in Information Systems *Graduated: 2026*

- GPA: 3.36/4.0 | IELTS: 7.0 | Interests: ML, time series forecasting, probabilistic forecasting, representation learning, graph algorithms, RL theory

Technical Skills

Languages: Python, SQL **ML/DL:** PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost, LightGBM
Time Series: ARIMA, SARIMA, ETS, state-space models, statsmodels, time-series CV, rolling-origin backtesting, look-ahead bias, data leakage, probabilistic forecasting, calibration
Data: NumPy, Pandas, PostgreSQL, MS SQL Server, MongoDB, DWH **NLP:** RAG, embeddings, FAISS, NLTK
CV/Research: OpenCV, SAM, CNN, GAN, VAE, TDA, Nilearn, NetworkX, giotto-tda
Tools: Git, GitHub, GitLab, Docker, Docker-compose, FastAPI, Flask, MLflow, Jupyter **BI:** Power BI, Matplotlib, Seaborn
ML: classification, regression, clustering, PCA/UMAP, meta-learning **Math:** probability, statistics, stochastic processes, linear algebra, calculus, algorithms

Research Projects

Thesis: Systematic Review and Meta-Analysis of RAG in Medicine Jun. 2026

- Conducted a systematic review and meta-analysis of RAG in medical AI as part of a 4-person team: 255 experimental results from 33 studies
- Analyzed the effect of model size, task type, and retrieval architecture; models <10B showed a +8.36 pp improvement, approaching the quality of significantly larger closed-source models
- Agentic and sparse retrieval yielded +29.72 pp versus +6.94 pp for dense retrieval; responsible for literature screening, architecture analysis, and the statistical/predictive component

Geometric Characterization of Reasoning via Topological Analysis of fMRI Dec. 2025

- Carried out a full fMRI research pipeline: preprocessing (Nilearn, FSL), persistent homology (TDA), and graph analysis of the neural correlates of logical reasoning
- Identified a negative correlation ($r = -0.83$) between neural trajectory speed and topological complexity; slower reasoning was associated with more complex neural dynamics
- Contributed to a 3D CNN pipeline achieving 68–70% cross-subject accuracy with Grad-CAM; Random Forest linked solving success to H0 persistence entropy and Small-Worldness ($R^2 = 0.17$)

Projects

Multimodal Search System for Industrial Spare Parts | GitHub Jun. 2024 – Aug. 2024

- Built an image/text search system for the JCB catalog with 100,000+ records; consolidated data from MS SQL Server and external sources into PostgreSQL
- Reconstructed relationships across thousands of tables and implemented bilingual semantic search using embeddings, FAISS, and FastAPI
- Reverse-engineered a legacy application for R&D visual search; adapted SAM for segmentation and prepared data for GAN-based augmentation

RAG System for Corporate Documents | *Freedom Hackathon* | GitHub Aug. 2024

- Built a pipeline covering document parsing, semantic chunking, dense retrieval, and constrained generation
- Implemented document-type detection and table extraction from PDF/Excel with Cyrillic support
- Found that retrieval quality depends more heavily on preprocessing and chunking strategy than on LLM choice

Sales Forecasting and Pattern Discovery Mar. 2024

- Analyzed 359,000+ sales records; performed time-series EDA, stationarity checks (ADF/KPSS), seasonal decomposition, and feature engineering with temporal, temperature, and tourism features (correlation: 0.623)
- Compared seasonal naive, ARIMA/SARIMA, ETS, and gradient boosting models (LightGBM/XGBoost/CatBoost) using rolling-origin backtesting and time-series cross-validation; reduced RMSE to 37–38
- Applied UMAP and DBSCAN for product segmentation; built interactive Power BI dashboards for sales and product-segment analysis